IIT HYDERABAD

Department of Electrical Engineering

Electronic Devices and Circuits Laboratory Report

Experiment No: **02**

Course Name: Electronic Devices and Circuits Lab

Course Code: EE2301

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Contents

1 Objective

- To plot the input and output characteristics of a BJT in CE, CB, and CC configurations.
- To determine the current gains (β, α, γ), input resistances (r_i), and output resistances (r_o) for each configuration.
- To experimentally verify the theoretical relations: $\beta = \frac{\alpha}{1-\alpha}$ and $\gamma = \beta + 1$.

2 Components Required

- BC547 transistor - 1
- Regulated DC Power Supply
- Resistors (1 kΩ, 10 kΩ)
- Voltmeter
- Ammeter (Micro,Milli)
- Breadboard
- Connecting wires

3 Part A – CE Configuration: Output Characteristics

3.1 Aim

To plot the output characteristics of a BJT in the Common Emitter (CE) configuration by plotting collector current I_C against collector-emitter voltage V_{CE} for different constant values of base current I_B, and to determine the current gain β and output resistance r_o.

3.2 Theory

In the Common Emitter configuration, the emitter terminal is common to both the input and output circuits.

The output characteristic represents the variation of collector current I_C with collector-emitter voltage V_{CE} for a constant base current I_B:

$$I_C = f(V_{CE})\Big|_{I_B=\text{constant}}.$$

In the active region of operation, the collector current is approximately proportional to the base current:

$$I_C \approx \beta I_B.$$

The current gain is therefore

$$\boxed{\beta = \frac{\Delta I_C}{\Delta I_B}}$$

The output resistance is determined from the slope of the output characteristic:

$$\boxed{r_o = \frac{\Delta V_{CE}}{\Delta I_C}}$$

for constant I_B.

The CE output characteristic consists of a saturation region at low V_{CE} followed by an approximately flat active region. The small positive slope in the active region is due to the Early effect and corresponds to a finite output resistance.

3.3 Circuit

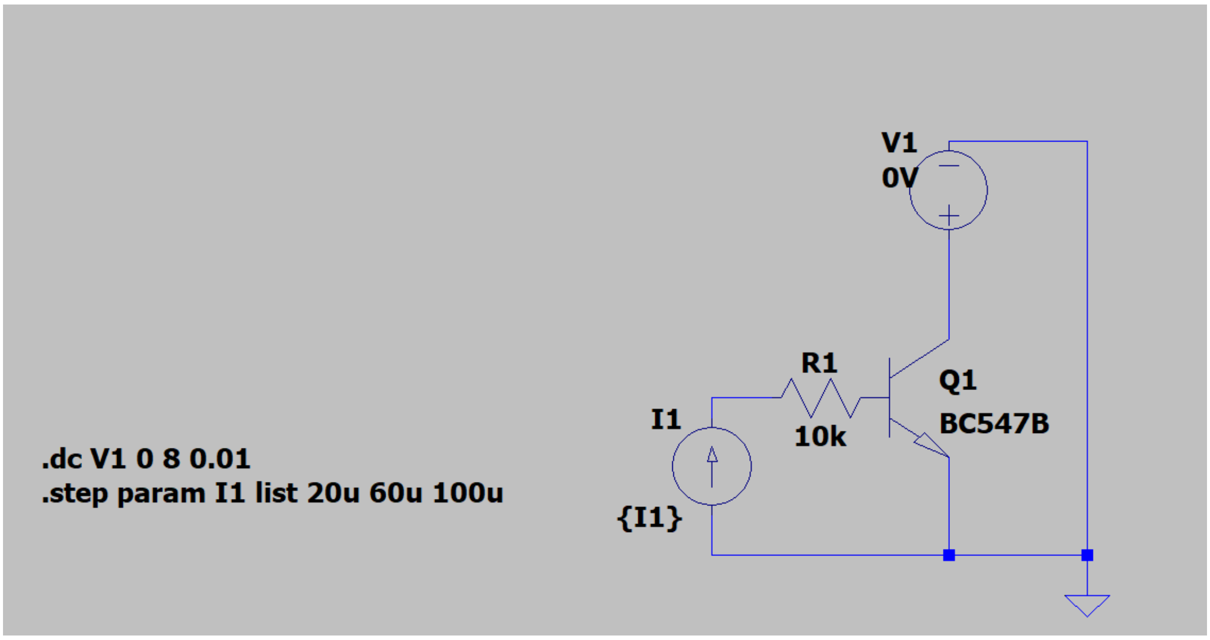


Figure 1: circuit

3.4 Observation Table

Table 1: CE Output Characteristics

S.No.	V_{CE} **(V)**	$I_B = 20$ I_C **(mA)**	$I_B = 40$ I_C **(mA)**	$I_B = 60$ I_C **(mA)**	$I_B = 100$ I_C **(mA)**
1	0.0	0.0035	0.0037	0.0030	0.003
2	0.5	6.76	11.96	16.39	22.50
3	1.0	6.78	13.17	19.19	27.53
4	1.5	6.86	13.38	19.60	30.65
5	2.0	6.91	13.56	19.79	31.59
6	4.0	7.18	13.90	20.67	33.27
7	6.0	7.12	14.30	21.30	34.61
8	8.0	7.50	14.89	22.01	35.56
9	10.0	7.60	15.40	–	–
10	12.0	7.80	15.71	–	–
11	14.0	7.91	15.96	–	–
12	16.0	8.05	16.26	–	–
13	18.0	8.20	17.10	–	–
14	20.0	8.40	–	–	–

Note: Blank entries indicate readings that were not clearly available in the provided laboratory record.

3.5 Graph

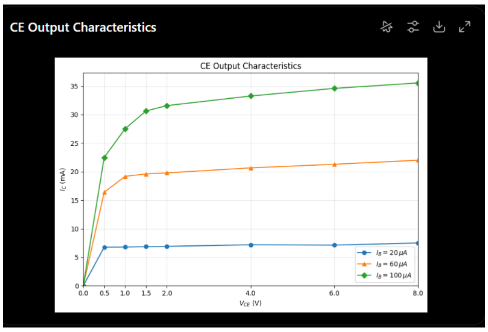


Figure 2: python graph (experimental)

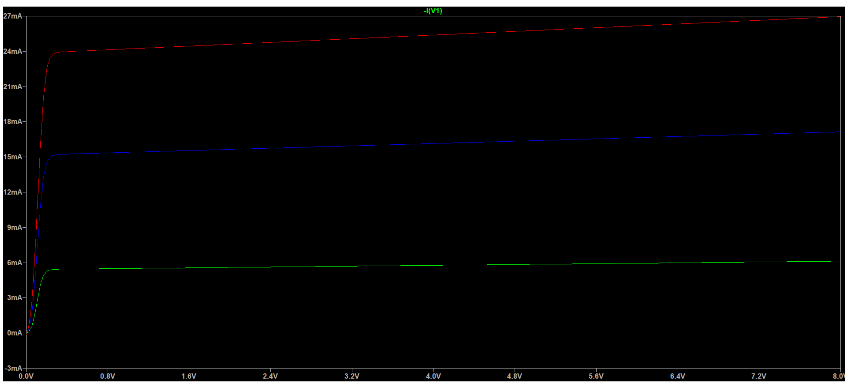


Figure 3: Simulation (green for 0.02 mA, Blue for 0.06 mA, Red for 0.1 mA)

3.6 Calculations

3.6.1 Current Gain β

The current gain of the BJT in the Common Emitter configuration is given by

$$\beta = \frac{\Delta I_C}{\Delta I_B}.$$

For comparison, the collector currents at the same collector-emitter voltage of $V_{CE} = 8\,\mathrm{V}$ are taken from the measured output characteristics:

I_B (μA)	I_C (mA)
20	7.50
60	22.01
100	35.56

Using the first and last measured points,

$$\Delta I_C = 35.56 - 7.50 = 28.06\,\mathrm{mA}$$

and

$$\Delta I_B = 100 - 20 = 80\,\mu\mathrm{A} = 0.080\,\mathrm{mA}.$$

Therefore,

$$\beta = \frac{28.06}{0.080}$$

$$\boxed{\beta_{\mathrm{exp}} \approx 350.75}$$

Thus, the experimental current gain of the transistor in the CE configuration is

$$\boxed{\beta_{\mathrm{exp}} \approx 351}$$

3.6.2 Output Resistance r_o

The output resistance is obtained from the slope of the CE output characteristic at constant base current:

$$r_o = \frac{\Delta V_{CE}}{\Delta I_C}.$$

Using the $I_B = 20\,\mu\mathrm{A}$ characteristic between $V_{CE} = 4\,\mathrm{V}$ and $V_{CE} = 8\,\mathrm{V}$:

$$I_C(4\,\mathrm{V}) = 7.18\,\mathrm{mA}$$

and

$$I_C(8\,\mathrm{V}) = 7.50\,\mathrm{mA}.$$

Therefore,

$$\Delta V_{CE} = 8 - 4 = 4\,\mathrm{V}$$

and

$$\Delta I_C = 7.50 - 7.18 = 0.32\,\mathrm{mA}.$$

Converting the current difference into amperes,

$$0.32\,\mathrm{mA} = 0.32 \times 10^{-3}\,\mathrm{A}.$$

Hence,

$$r_o = \frac{4}{0.32 \times 10^{-3}}$$

$$r_o = 12500\,\Omega$$

and therefore,

$$\boxed{r_o \approx 12.5\,\mathrm{k}\Omega}$$

3.7 Result

The CE output characteristics were plotted for

$$I_B = 20\,\mu\mathrm{A}, \quad 60\,\mu\mathrm{A}, \quad 100\,\mu\mathrm{A}.$$

The experimentally determined current gain is

$$\boxed{\beta_{\mathrm{exp}} \approx 350.75}$$

and the output resistance obtained from the $I_B = 20\,\mu\mathrm{A}$ characteristic is

$$\boxed{r_o \approx 12.5\,\mathrm{k}\Omega}.$$

3.8 Observations

- The collector current increases significantly with increase in base current.
- At low values of V_{CE}, the transistor shows the saturation region.
- Beyond the saturation region, the collector current changes only gradually with V_{CE}.
- The slight positive slope of the active-region curves indicates a finite output resistance.

4 Part B – CE Configuration: Input Characteristics

4.1 Aim

To plot the input characteristics of a BJT in the Common Emitter configuration by plotting base current I_B against base-emitter voltage V_{BE} for different constant values of collector-emitter voltage V_{CE}, and to determine the input resistance r_i.

4.2 Theory

The CE input characteristic represents the variation of base current I_B with base-emitter voltage V_{BE} for a constant value of V_{CE}:

$$I_B = f(V_{BE})\Big|_{V_{CE}=\text{constant}}.$$

The base-emitter junction behaves approximately as a forward-biased PN junction. Hence, the base current is very small at low values of V_{BE} and increases rapidly after the knee region.

The input resistance is given by

$$\boxed{r_i = \frac{\Delta V_{BE}}{\Delta I_B}}$$

at constant V_{CE}.

4.3 Circuit

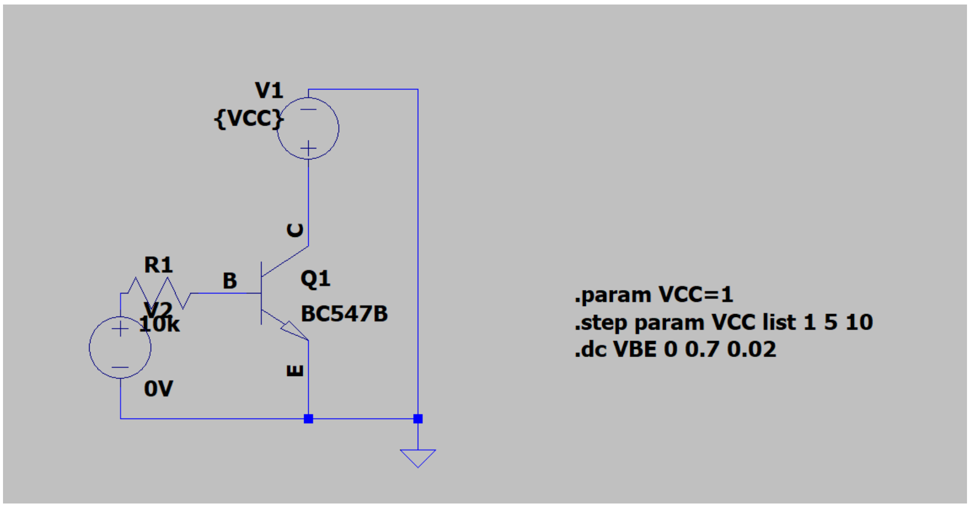


Figure 4: circuit

4.4 Observation Table

Table 2: CE Input Characteristics

S.No.	V_{BE} **(V)**	I_B **at 1 V (mA)**	I_B **at 5 V (mA)**	I_B **at 10 V (mA)**
1	0.00	0	0	0
2	0.05	0	0.0001	0
3	0.10	0	0.0001	0
4	0.15	0	0.0001	0
5	0.20	0	0.0001	0
6	0.25	0	0.0001	0
7	0.30	0	0.0001	0
8	0.35	0	0.0001	0
9	0.40	0	0.0001	0
10	0.42	0	0.0001	0
11	0.44	0	0.0001	0
12	0.46	0.0001	0.0001	0
13	0.48	0.0001	0.0001	0
14	0.50	0.0001	0.0001	0
15	0.52	0.0001	0.0001	0
16	0.54	0.0001	0.0001	0
17	0.56	0.0002	0.0002	0.0001
18	0.58	0.0004	0.0004	0.0003
19	0.60	0.0006	0.0009	0.0009
20	0.62	0.0017	0.0016	0.0018
21	0.64	0.0035	0.0037	0.0039
22	0.66	0.0073	0.0080	0.0130
23	0.68	0.0150	0.0200	–
24	0.70	0.0324	0.0565	–

Note: Current values are recorded in mA as written in the laboratory notebook. Therefore, for example,

$$0.001\,\text{mA} = 1\,\mu\text{A}.$$

4.5 Graph

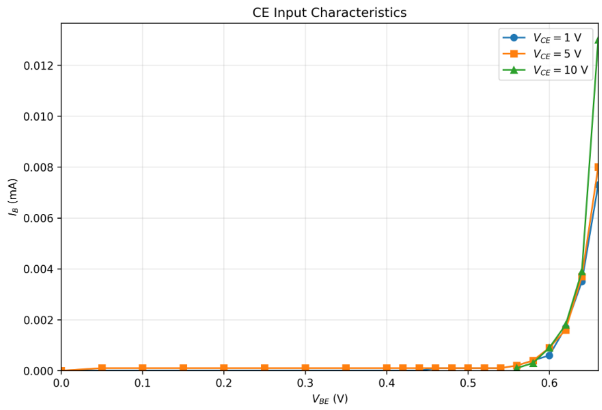


Figure 5: python graph (experimental)

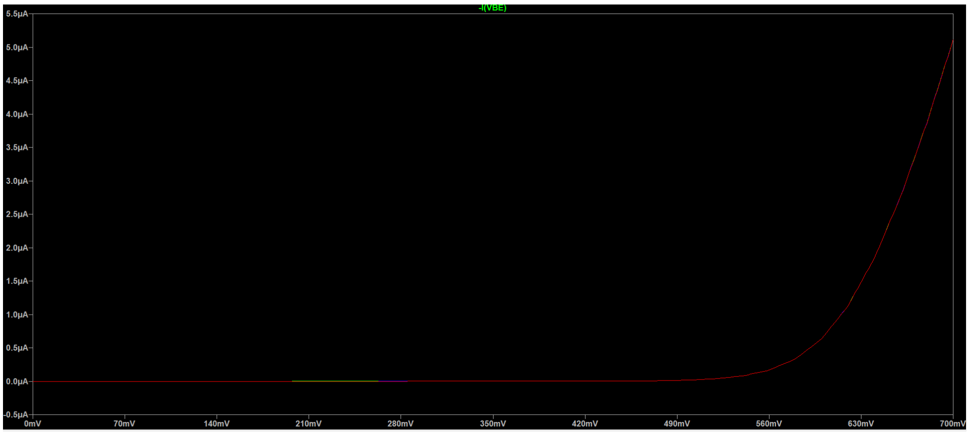


Figure 6: Simulation (all 3 almost got merged)

4.6 Calculations

4.6.1 Input Resistance r_i

The input resistance of the BJT in the Common Emitter configuration is obtained from the slope of the input characteristic:

$$r_i = \frac{\Delta V_{BE}}{\Delta I_B}.$$

The resistance is calculated using the measured values between $V_{BE} = 0.64\,\text{V}$ and $V_{BE} = 0.66\,\text{V}$.

For $V_{CE} = 1\,\text{V}$ From the observation table,

$$I_B(0.64\,\text{V}) = 0.0035\,\text{mA}$$

and

$$I_B(0.66\,\text{V}) = 0.0073\,\text{mA}.$$

Therefore,

$$\Delta V_{BE} = 0.66 - 0.64 = 0.02\,\text{V}$$

and

$$\Delta I_B = 0.0073 - 0.0035 = 0.0038\,\text{mA}.$$

Converting the current difference into amperes,

$$0.0038\,\text{mA} = 3.8 \times 10^{-6}\,\text{A}.$$

Hence,

$$r_i = \frac{0.02}{3.8 \times 10^{-6}}$$

$$\boxed{r_i \approx 5.26\,\text{k}\Omega}$$

For $V_{CE} = 5\,\text{V}$ From the observation table,

$$I_B(0.64\,\text{V}) = 0.0037\,\text{mA}$$

and

$$I_B(0.66\,\text{V}) = 0.0080\,\text{mA}.$$

Therefore,

$$\Delta I_B = 0.0080 - 0.0037 = 0.0043\,\text{mA}.$$

Hence,

$$r_i = \frac{0.02}{0.0043 \times 10^{-3}}$$

$$\boxed{r_i \approx 4.65\,\text{k}\Omega}$$

For $V_{CE} = 10\,\text{V}$ From the observation table,

$$I_B(0.64\,\mathrm{V}) = 0.0039\,\mathrm{mA}$$

and

$$I_B(0.66\,\mathrm{V}) = 0.0130\,\mathrm{mA}.$$

Therefore,

$$\Delta I_B = 0.0130 - 0.0039 = 0.0091\,\mathrm{mA}.$$

Hence,

$$r_i = \frac{0.02}{0.0091 \times 10^{-3}}$$

$$\boxed{r_i \approx 2.20\,\mathrm{k\Omega}}$$

The calculated input resistance values are summarized below.

Table 3: Calculated CE Input Resistance

V_{CE}	r_i
1 V	5.26 kΩ
5 V	4.65 kΩ
10 V	2.20 kΩ

4.7 Result

The CE input characteristics were plotted for

$$V_{CE} = 1\,\mathrm{V}, \quad 5\,\mathrm{V}, \quad 10\,\mathrm{V}.$$

The input resistance obtained from the measured characteristics is

$$\boxed{r_i \approx 5.26\,\mathrm{k\Omega} \quad (V_{CE} = 1\,\mathrm{V})}$$

$$\boxed{r_i \approx 4.65\,\mathrm{k\Omega} \quad (V_{CE} = 5\,\mathrm{V})}$$

and

$$\boxed{r_i \approx 2.20\,\mathrm{k\Omega} \quad (V_{CE} = 10\,\mathrm{V})}.$$

4.8 Observations

- The base current is very small at low values of V_{BE}.
- The base current starts increasing rapidly as the base-emitter junction becomes forward biased.
- The characteristic resembles the forward characteristic of a PN junction.
- The input resistance decreases as the base current increases.

5 Part C – CB Configuration: Output Characteristics

5.1 Aim

To plot the output characteristics of a BJT in the Common Base (CB) configuration by plotting collector current I_C against collector-base voltage V_{CB} for different constant values of emitter current I_E, and to determine the current gain α and output resistance r_o.

5.2 Theory

In the Common Base configuration, the base terminal is common to both the input and output circuits.

The output characteristic is given by

$$I_C = f(V_{CB})\Big|_{I_E=\text{constant}}.$$

The current gain in CB configuration is

$$\boxed{\alpha = \frac{\Delta I_C}{\Delta I_E}}$$

Since

$$I_E = I_C + I_B,$$

the collector current is slightly smaller than the emitter current and therefore

$$\alpha < 1.$$

The output resistance is

$$\boxed{r_o = \frac{\Delta V_{CB}}{\Delta I_C}}$$

at constant emitter current.

The CB output characteristics are nearly horizontal because the collector current is almost independent of the collector-base voltage in the active region.

5.3 Circuit

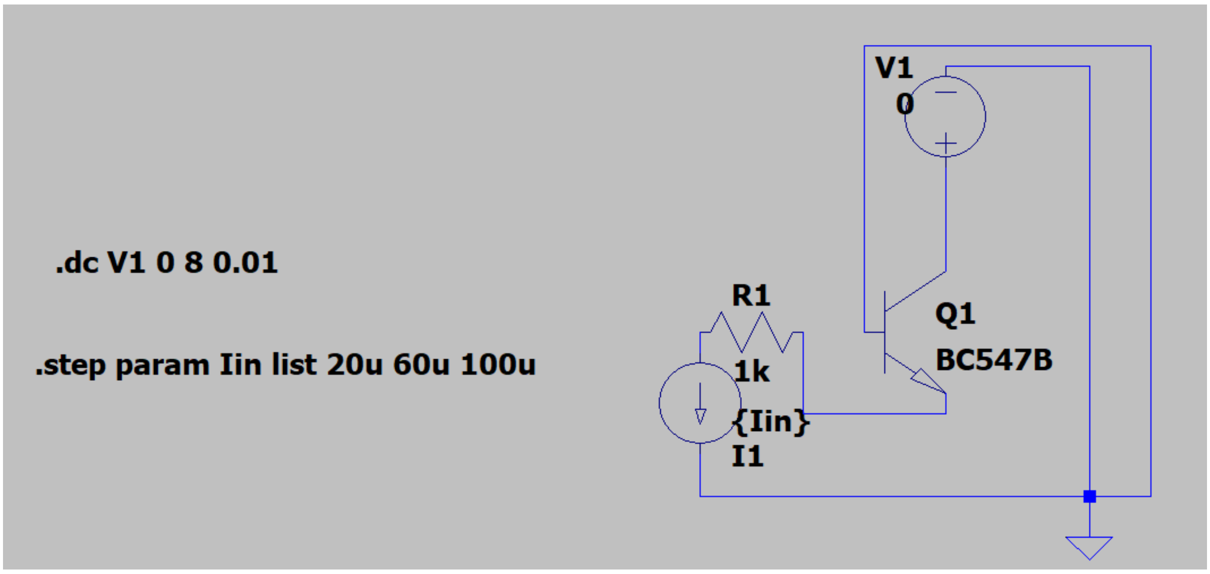


Figure 7: circuit

5.4 Observation Table

Table 4: CB Output Characteristics

S.No.	V_{CB} (V)	$I_E = 2$ mA I_C (mA)	$I_E = 6$ mA I_C (mA)	$I_E = 10$ mA I_C (mA)
1	0.0	1.990	5.987	9.965
2	0.5	1.990	5.987	9.965
3	1.0	1.990	5.987	9.965
4	1.5	1.990	5.990	9.965
5	2.0	1.991	5.990	9.970
6	4.0	1.992	5.990	9.970
7	6.0	1.994	5.990	9.970
8	8.0	1.996	5.990	9.970
9	10.0	1.998	–	–
10	12.0	1.999	–	–

Note: Only values that could be read with reasonable confidence from the laboratory notebook have been entered.

5.5 Graph

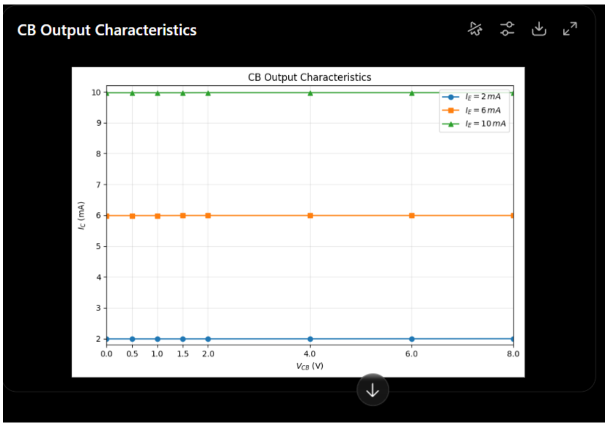


Figure 8: python graph (expiremental)

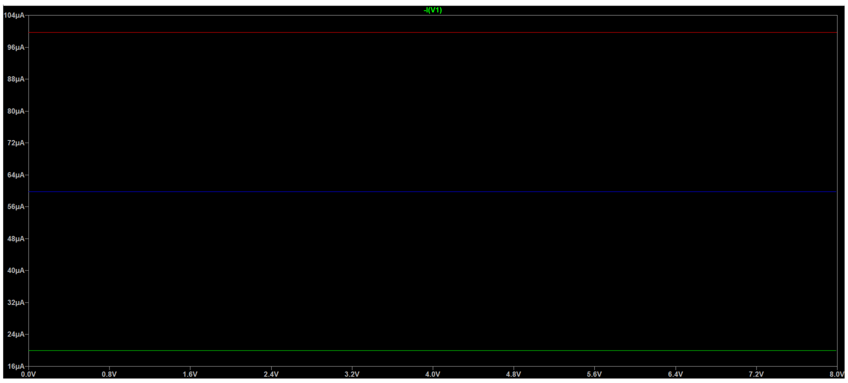


Figure 9: simulation(green for 2 mA,Blue for 6 mA,Red for 10 mA

5.6 Calculations

5.6.1 Current Gain α

The current gain of the BJT in the Common Base configuration is given by

$$\alpha = \frac{\Delta I_C}{\Delta I_E}.$$

For a consistent comparison, the collector currents at the same collector-base voltage of $V_{CB} = 2\,\mathrm{V}$ are taken from the measured CB output characteristics:

I_E (mA)	I_C (mA)
2	1.991
6	5.990
10	9.970

Using the first and last points,

$$\Delta I_C = 9.970 - 1.991 = 7.979\,\text{mA}$$

and

$$\Delta I_E = 10 - 2 = 8\,\text{mA}.$$

Therefore,

$$\alpha = \frac{7.979}{8}$$

$$\boxed{\alpha_{\text{exp}} \approx 0.9974}$$

The value is slightly less than unity, as expected for a transistor operating in the Common Base configuration.

5.6.2 Output Resistance r_o

The output resistance is obtained from the slope of the CB output characteristic at constant emitter current:

$$r_o = \frac{\Delta V_{CB}}{\Delta I_C}.$$

Using the $I_E = 2\,\text{mA}$ characteristic between $V_{CB} = 2\,\text{V}$ and $V_{CB} = 8\,\text{V}$:

$$I_C(2\,\text{V}) = 1.991\,\text{mA}$$

and

$$I_C(8\,\text{V}) = 1.996\,\text{mA}.$$

Therefore,

$$\Delta V_{CB} = 8 - 2 = 6\,\text{V}$$

and

$$\Delta I_C = 1.996 - 1.991 = 0.005\,\text{mA}.$$

Converting the current difference into amperes,

$$0.005\,\text{mA} = 5 \times 10^{-6}\,\text{A}.$$

Hence,

$$r_o = \frac{6}{5 \times 10^{-6}}$$

$$r_o = 1.2 \times 10^{6}\,\Omega$$

and therefore,

$$\boxed{r_o \approx 1.20\,\text{M}\Omega}$$

5.7 Result

The CB output characteristics were plotted for

$$I_E = 2\,\text{mA}, \quad 6\,\text{mA}, \quad 10\,\text{mA}.$$

The experimentally determined common-base current gain is

$$\boxed{\alpha_{\text{exp}} \approx 0.9974}$$

and the output resistance obtained from the $I_E = 2\,\text{mA}$ characteristic is

$$\boxed{r_o \approx 1.20\,\text{M}\Omega}.$$

5.8 Observations

- The collector current is slightly smaller than the emitter current.
- The current gain α is close to unity.
- The collector current changes only slightly with increase in V_{CB}.
- The CB output characteristics are nearly horizontal.
- The small slope indicates a high output resistance.

6 PartD – CB Input Characteristics

6.1 Aim

To plot the input characteristics of a BJT in the Common Base (CB) configuration and to determine the input resistance r_i.

6.2 Circuit Diagram

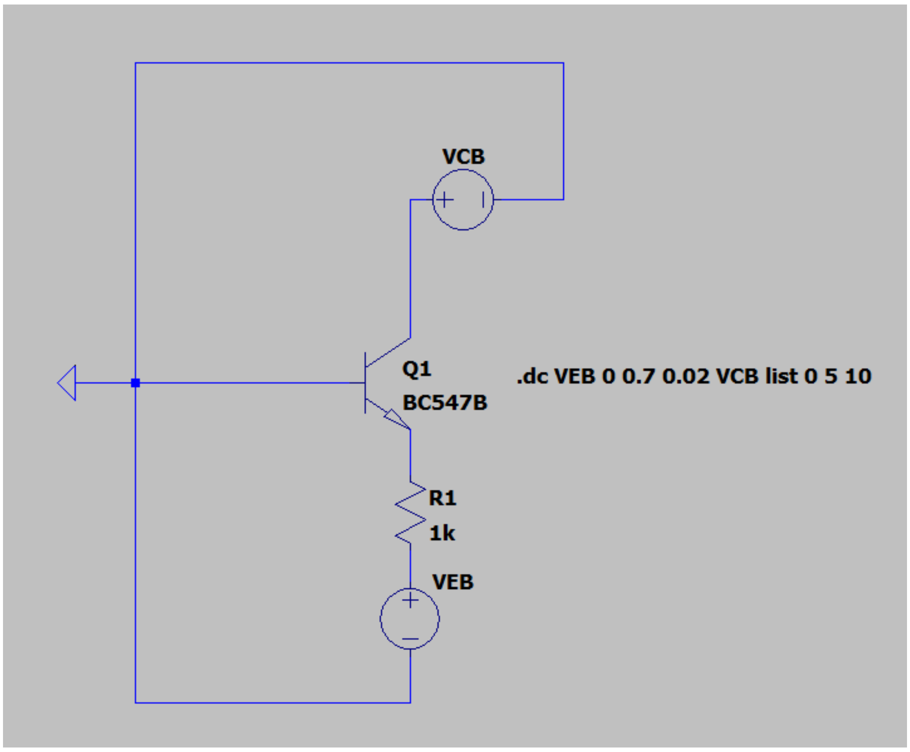


Figure 10: LT Spice Circuit Schematic

6.3 Theory

In the Common Base (CB) configuration, the base terminal is common to both the input and output circuits. The input is applied between the emitter and base, while the output is taken between the collector and base.

The CB input characteristic represents the relationship between the emitter current I_E and the emitter-base voltage V_{EB} at a constant collector-base voltage V_{CB}. The characteristic is diode-like, since the emitter-base junction is forward biased.

As V_{EB} increases, the emitter current I_E increases rapidly. The input resistance of the CB configuration is determined from the slope of the input characteristic and is given by

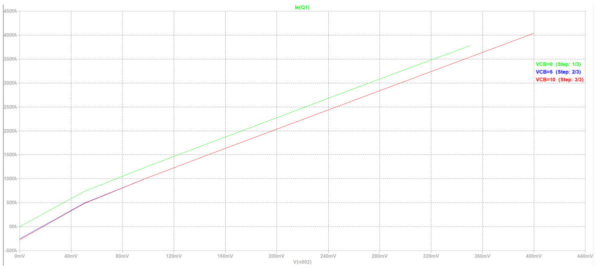
$$\mathbf{r_i} = \left.\frac{\mathbf{\Delta V_{EB}}}{\mathbf{\Delta I_E}}\right|_{\mathbf{V_{CB}}=\text{constant}}$$

The CB configuration has a low input resistance. The input characteristics are obtained for different values of V_{CB}, such as 0 V, 5 V, and 10 V.

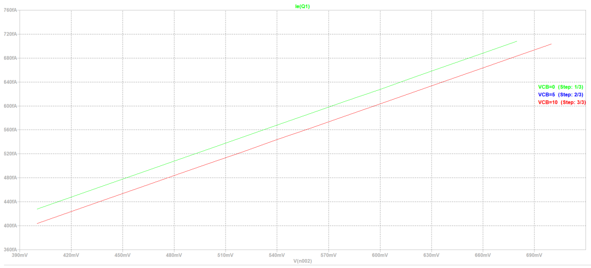
6.4 Observation Table

V_{EB} (V)	I_E (mA) at $V_{CB} = 0\,\text{V}$	I_E (mA) at $V_{CB} = 5\,\text{V}$	I_E (mA) at $V_{CB} = 10\,\text{V}$
0.00	0	0	0
0.05	0	0	0
0.10	0	0	0
0.15	0	0	0
0.20	0	0	0
0.25	0	0	0
0.30	0	0	0
0.35	0	0	0
0.40	0.0002	0.0001	0.0001
0.42	0.0002	0.0002	0.0002
0.44	0.0004	0.0005	0.0004
0.46	0.0010	0.0010	0.0011
0.48	0.0020	0.0022	0.0023
0.50	0.0043	0.0052	0.0055
0.52	0.0108	0.0115	0.0113
0.54	0.0230	0.0250	0.0250
0.56	0.0505	0.0560	0.0550
0.58	0.1050	0.1165	0.1150
0.60	0.2270	0.2560	0.2510
0.62	0.4880	0.5680	0.5800
0.64	1.1250	–	–
0.66	2.7300	–	–

6.5 Plot



(a) Low Voltage Range ($V_{EB} = 0 - 0.4$)



(b) Conduction Range ($V_{EB} = 0.4 - 0.7$)

Figure 11: CB Characterstics (I_E vs V_{EB}) for various V_{CB} values

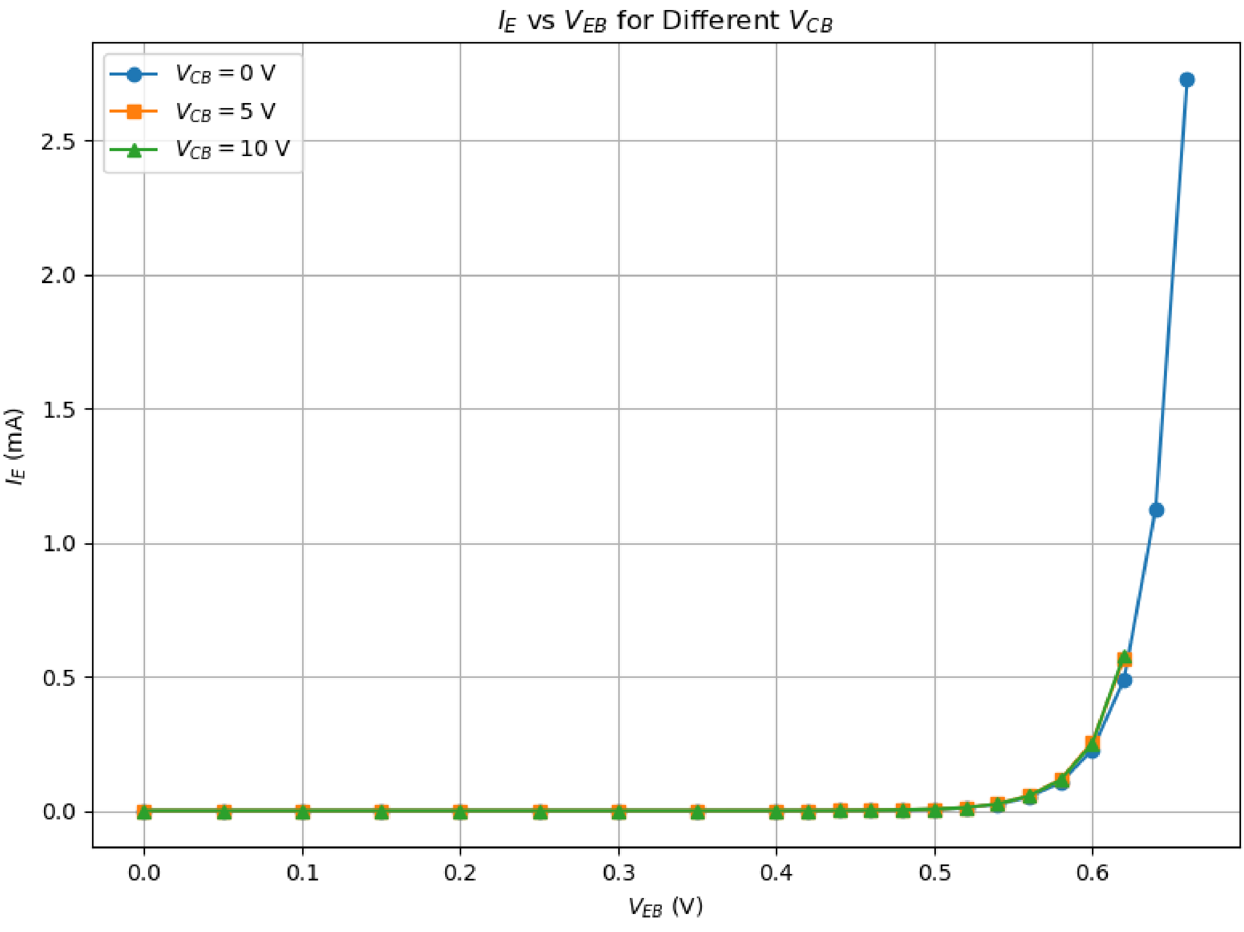


Figure 12: Experimental CB Input Characterstics I_E vs V_{EB}

6.6 Calculations

6.6.1 Input resistance r_i

Input resistance (r_i) is the ratio of the change in input voltage to the corresponding change in input current.

$$r_i = \left.\frac{\Delta V_{EB}}{\Delta I_E}\right|_{V_{CB}=\text{const}}$$

At $V_{CB} = 0\,\text{V}$:

$$\Delta V_{EB} = 0.56 - 0.54 = 0.02\,\text{V}$$

$$\Delta I_E = 0.0505\,\text{mA} - 0.0230\,\text{mA} = 0.0275\,\text{mA}$$

$$r_{i1} = \frac{0.02\,\text{V}}{0.0275\,\text{mA}} = \mathbf{727.27\,\Omega}$$

At $V_{CB} = 5\,\text{V}$:

$$\Delta V_{EB} = 0.56 - 0.54 = 0.02\,\text{V}$$

$$\Delta I_E = 0.0560\,\text{mA} - 0.0250\,\text{mA} = 0.0310\,\text{mA}$$

$$r_{i2} = \frac{0.02\,\text{V}}{0.0310\,\text{mA}} = \mathbf{645.16\,\Omega}$$

At $V_{CB} = 10\,\text{V}$:

$$\Delta V_{EB} = 0.56 - 0.54 = 0.02\,\text{V}$$

$$\Delta I_E = 0.0550\,\text{mA} - 0.0250\,\text{mA} = 0.0300\,\text{mA}$$

$$r_{i3} = \frac{0.02\,\text{V}}{0.0300\,\text{mA}} = \mathbf{666.67\,\Omega}$$

6.7 Result

The CB input characteristics were plotted for $V_{CB} = 0\,\text{V}$, $5\,\text{V}$, and $10\,\text{V}$.

The input resistance obtained from the measured characteristics is

$$r_i \approx 727.27\,\Omega \quad (V_{CB} = 0\,\text{V}),$$

$$r_i \approx 645.16\,\Omega \quad (V_{CB} = 5\,\text{V}),$$

and

$$r_i \approx 666.67\,\Omega \quad (V_{CB} = 10\,\text{V}).$$

6.8 Observations

- The emitter current I_E increases rapidly with increase in emitter-base voltage V_{EB}.
- The input characteristic exhibits diode-like behaviour due to the forward-biased emitter-base junction.
- The CB configuration has a relatively low input resistance.
- The characteristics for different values of V_{CB} are closely spaced in the conduction region.

7 Part E – CC Output Characteristics

7.1 Aim

To study the output characteristics of a BJT in the Common Collector (CC) configuration and determine the current gain γ and output dynamic resistance r_o.

7.2 Circuit

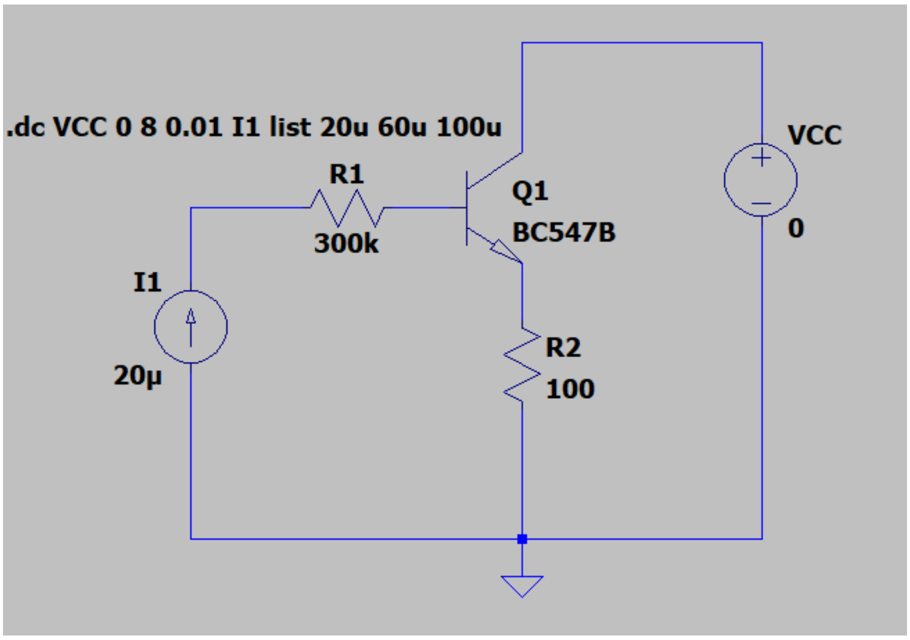


Figure 13: LTspice Schematic of CC output characteristics

7.3 Theory

In a Common Collector (CC) configuration, the collector terminal is common to both the input and output circuits. The output characteristic represents the variation of the emitter current I_E with the collector-emitter voltage V_{CE} for a constant base current I_B.

For a fixed value of I_B, the emitter current I_E remains nearly constant as V_{CE} is increased in the active region. Hence, the output characteristics are nearly horizontal, with a slight increase in I_E due to the Early effect.

The output resistance of the CC configuration is given by

$$r_o = \left(\frac{\Delta V_{CE}}{\Delta I_E}\right)_{I_B=\text{constant}}$$

where r_o is the small-signal output resistance.

7.4 Observation Table

S.No.	V_{CE} (V)	I_E (mA) at $I_B = 20\,\mu A$	I_E (mA) at $I_B = 60\,\mu A$	I_E (mA) at $I_B = 90\,\mu A$
1	0.0	0.017	0.051	0.092
2	0.5	6.210	15.97	21.08
3	1.0	6.245	18.20	24.65
4	1.5	6.300	18.56	26.55
5	2.0	6.329	18.87	27.24
6	4.0	6.470	19.64	28.90
7	6.0	6.600	21.08	30.55
8	8.0	6.742	21.89	31.88

7.5 Simulation plot

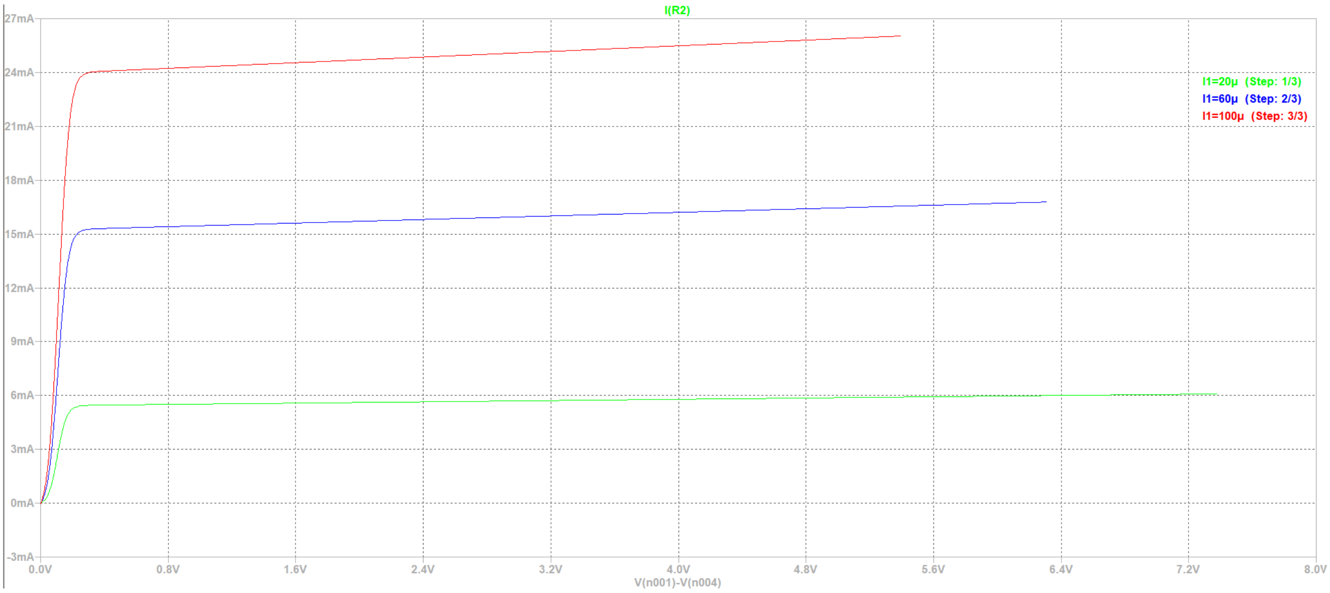
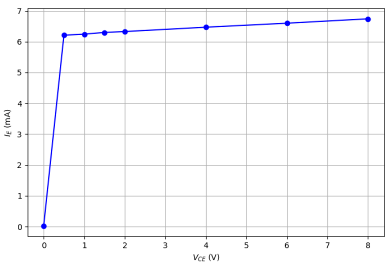
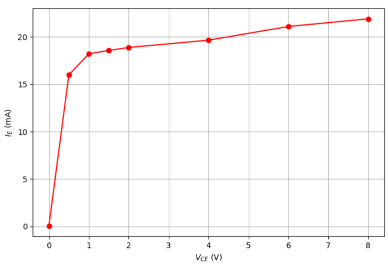


Figure 14: LTspice Simulation of CC output characteristics I_E vs V_{CE} for $I_B = 20\,\mu\text{A},\ 60\,\mu\text{A},$ and $100\,\mu\text{A}$

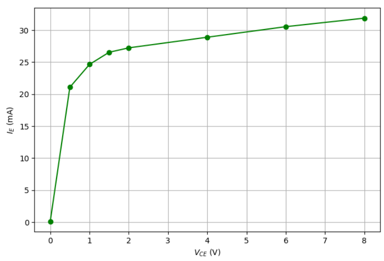
7.6 Experimental Characteristic Plots



(a) $I_B = 20\,\mu\text{A}$



(b) $I_B = 60\,\mu\text{A}$



(c) $I_B = 90\,\mu\text{A}$

Figure 15: Experimental plots of CC Output characteristics (I_E vs V_{CE})

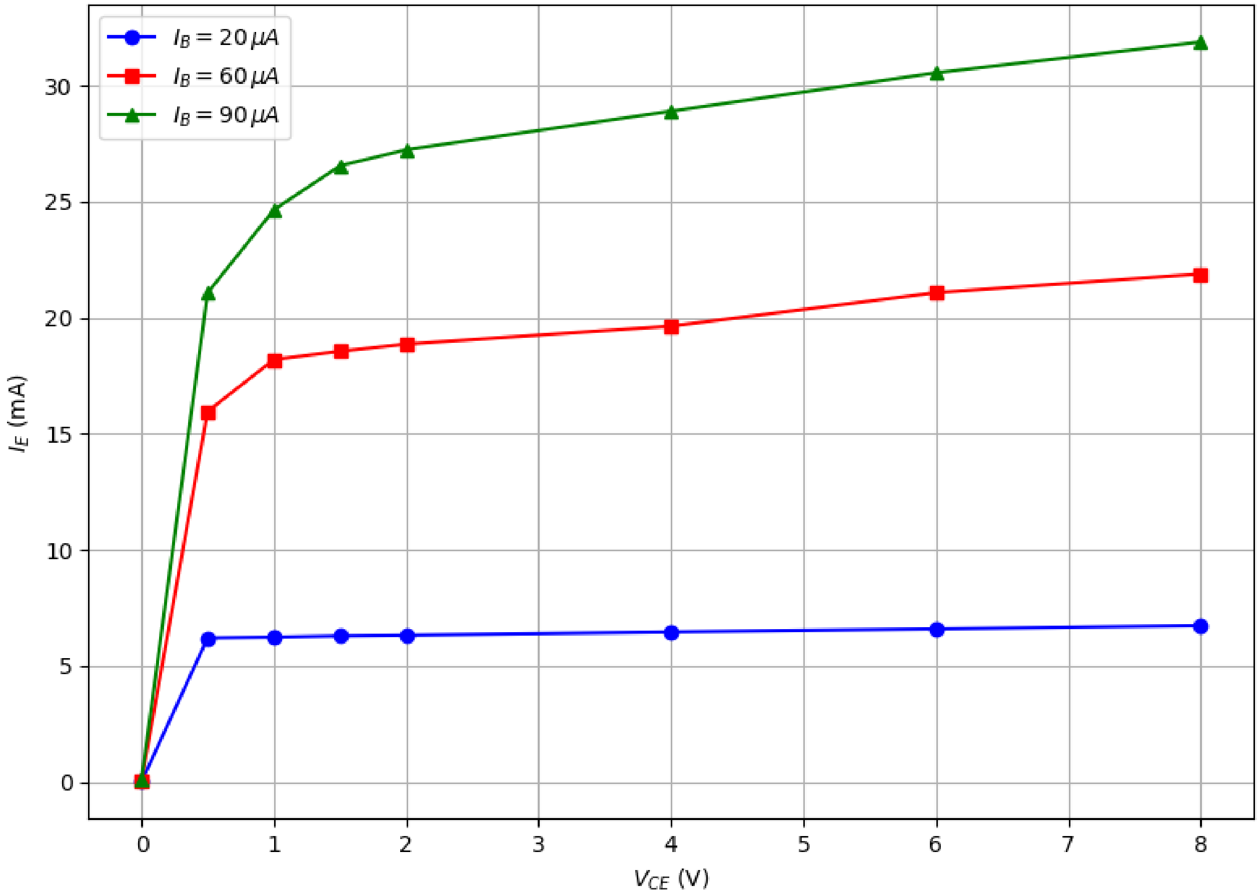


Figure 16: CC Output Characteristics I_E vs V_{CE}

7.7 Calculations

7.7.1 Current Gain

The current gain γ of the Common Collector (CC) configuration is given by

$$\gamma = \frac{\Delta I_E}{\Delta I_B}$$

Using the values of I_E corresponding to $I_B = 20\,\mu\text{A}$ and $I_B = 90\,\mu\text{A}$,

$$\Delta I_B = 90\,\mu\text{A} - 20\,\mu\text{A} = 70\,\mu\text{A} = 0.070\,\text{mA}$$

At $V_{CE} = 4\,\text{V}$:

$$\gamma_4 = \frac{28.90\,\text{mA} - 6.470\,\text{mA}}{0.070\,\text{mA}} = \mathbf{320.43}$$

At $V_{CE} = 6\,\text{V}$:

$$\gamma_6 = \frac{30.55\,\text{mA} - 6.600\,\text{mA}}{0.070\,\text{mA}} = \mathbf{342.14}$$

At $V_{CE} = 8\,\text{V}$:

$$\gamma_8 = \frac{31.88\,\text{mA} - 6.742\,\text{mA}}{0.070\,\text{mA}} = \mathbf{359.11}$$

7.7.2 Average Current Gain

The average current gain is calculated as

$$\gamma_{\text{avg}} = \frac{\gamma_4 + \gamma_6 + \gamma_8}{3}$$

$$\gamma_{\text{avg}} = \frac{320.43 + 342.14 + 359.11}{3}$$

$$\gamma_{\text{avg}} = 340.56$$

7.7.3 Output Resistance r_o

The output resistance r_o is calculated from the slope of the output characteristic in the good (active) region:

$$r_o = \left(\frac{\Delta V_{CE}}{\Delta I_E}\right)_{I_B=\text{constant}}$$

Taking the good region from $V_{CE} = 4\,\text{V}$ to $V_{CE} = 8\,\text{V}$:

At $I_B = 20\,\mu\text{A}$:

$$r_o = \frac{8\,\text{V} - 4\,\text{V}}{6.742\,\text{mA} - 6.470\,\text{mA}} = \frac{4}{0.272} = 14.71\,\text{k}\Omega$$

At $I_B = 60\,\mu\text{A}$:

$$r_o = \frac{8\,\text{V} - 4\,\text{V}}{21.89\,\text{mA} - 19.64\,\text{mA}} = \frac{4}{2.25} = 1.78\,\text{k}\Omega$$

At $I_B = 90\,\mu\text{A}$:

$$r_o = \frac{8\,\text{V} - 4\,\text{V}}{31.88\,\text{mA} - 28.90\,\text{mA}} = \frac{4}{2.98} = 1.34\,\text{k}\Omega$$

7.8 Result

The CC output characteristics were plotted for $I_B = 20\,\mu\text{A}$, $60\,\mu\text{A}$, and $90\,\mu\text{A}$.

The experimentally determined current gains are

$$\gamma_4 = 320.43, \qquad \gamma_6 = 342.14, \qquad \gamma_8 = 359.11.$$

The average current gain is

$$\gamma_{\text{avg}} = 340.56.$$

The output resistance obtained from the active-region characteristics is

$$r_o = 14.71\,\mathrm{k\Omega} \quad (I_B = 20\,\mu\mathrm{A}),$$

$$r_o = 1.78\,\mathrm{k\Omega} \quad (I_B = 60\,\mu\mathrm{A}),$$

and

$$r_o = 1.34\,\mathrm{k\Omega} \quad (I_B = 90\,\mu\mathrm{A}).$$

7.9 Observations

- The emitter current I_E increases with increase in base current I_B.
- For a fixed I_B, the emitter current remains nearly constant with increase in V_{CE} in the active region.
- The output characteristics are nearly horizontal in the active region.
- A slight positive slope is observed in the active region, indicating a finite output resistance.
- The current gain γ is significantly greater than unity.

8 Part F – CC Input Characteristics

8.1 Aim

To study the input characteristics of a BJT in the Common Collector (CC) configuration and determine the input dynamic resistance r_i for different values of V_{CE}.

8.2 Circuit

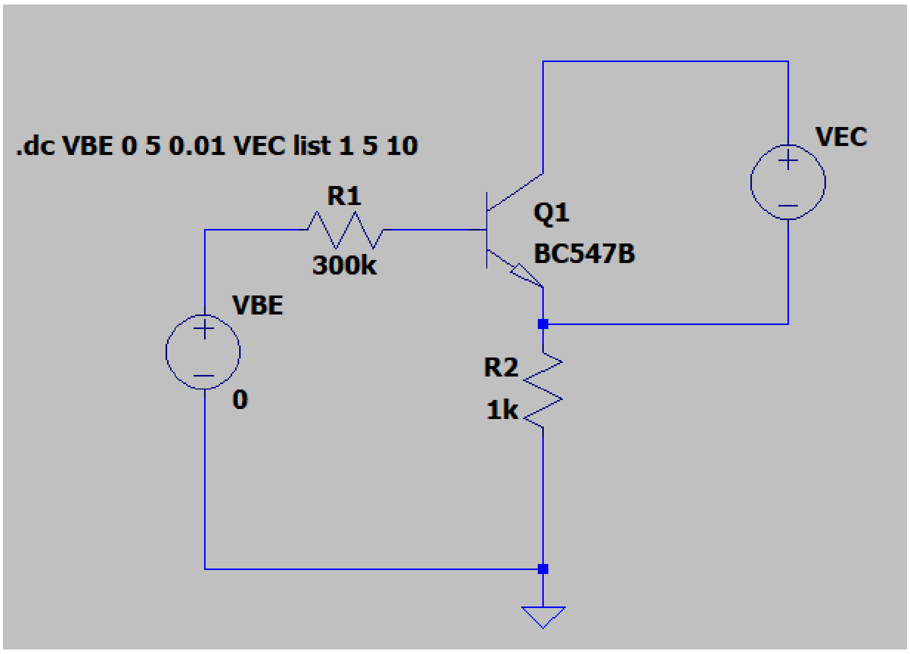


Figure 17: LTspice Schematic of CC Input Configuration

8.3 Theory

Theory:

In a Common Collector (CC) configuration, the collector terminal is common to both the input and output circuits. The input characteristic represents the variation of base current I_B with the collector-base voltage V_{CB} for a constant collector-emitter voltage V_{CE}.

Since

$$V_{CE} = V_{CB} + V_{BE},$$

for constant V_{CE},

$$V_{BE} = V_{CE} - V_{CB}.$$

As V_{CB} increases, V_{BE} decreases and consequently the base current I_B decreases. Hence, the input characteristic is diode-like and shows a rapid decrease in I_B with increasing V_{CB}.

The input resistance of the CC configuration is given by

$$r_i = \left(\frac{\Delta V_{BE}}{\Delta I_B}\right)_{V_{CE}=\text{constant}}.$$

Since V_{CE} is constant,

$$|\Delta V_{BE}| = |\Delta V_{CB}|,$$

and therefore the magnitude of the input resistance can also be obtained from

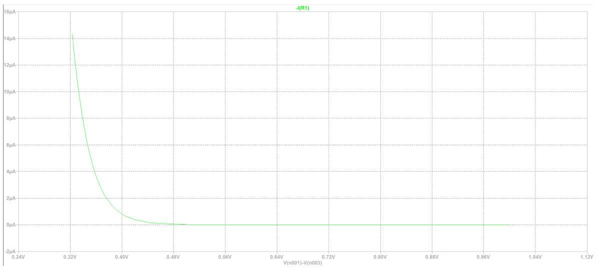
$$r_i = \left| \frac{\Delta V_{CB}}{\Delta I_B} \right|_{V_{CE}=\text{constant}} .$$

where r_i is the small-signal input resistance.

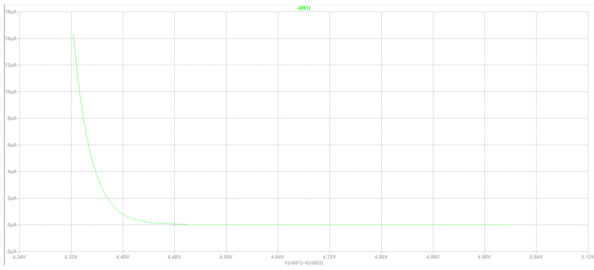
8.4 Observation Table

S.No.	$V_{CE} = 1\,\mathrm{V}$		$V_{CE} = 5\,\mathrm{V}$		$V_{CE} = 10\,\mathrm{V}$	
	V_{CB} (V)	I_B (µA)	V_{CB} (V)	I_B (µA)	V_{CB} (V)	I_B (µA)
1	0.26	94.5	4.30	61.6	9.34	13.7
2	0.27	72.8	4.32	23.5	9.36	5.4
3	0.29	30.5	4.34	9.4	9.38	2.5
4	0.31	22.0	4.36	4.7	9.40	1.0
5	0.33	11.7	4.38	2.5	9.41	0.4
6	0.35	6.4	4.40	1.0	9.42	0.0
7	0.36	4.7	4.42	0.3	9.50	0.0
8	0.37	3.6	4.44	0.1	–	–
9	0.38	3.0	4.47	0.0	–	–
10	0.39	1.5	4.50	0.0	–	–
11	0.41	1.0	4.85	0.0	–	–
12	0.42	0.7	–	–	–	–
13	0.53	0.1	–	–	–	–
14	0.55	0.0	–	–	–	–
15	0.90	0.0	–	–	–	–

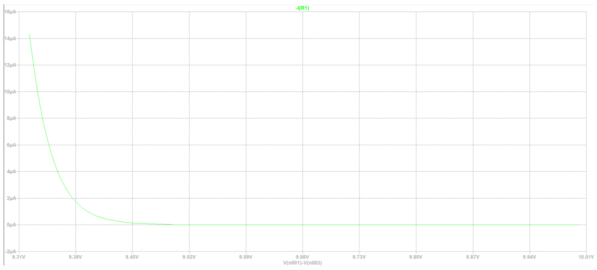
8.5 Simulation plots



(a) I_B vs V_{CB} at fixed $V_{CE} = 1$



(b) I_B vs V_{CB} at fixed $V_{CE} = 5$



(c) I_B vs V_{CB} at fixed $V_{CE} = 10$

Figure 18: LTspice Simulation of CC Input Characteristics (I_B vs V_{CB}) at different V_{CE}

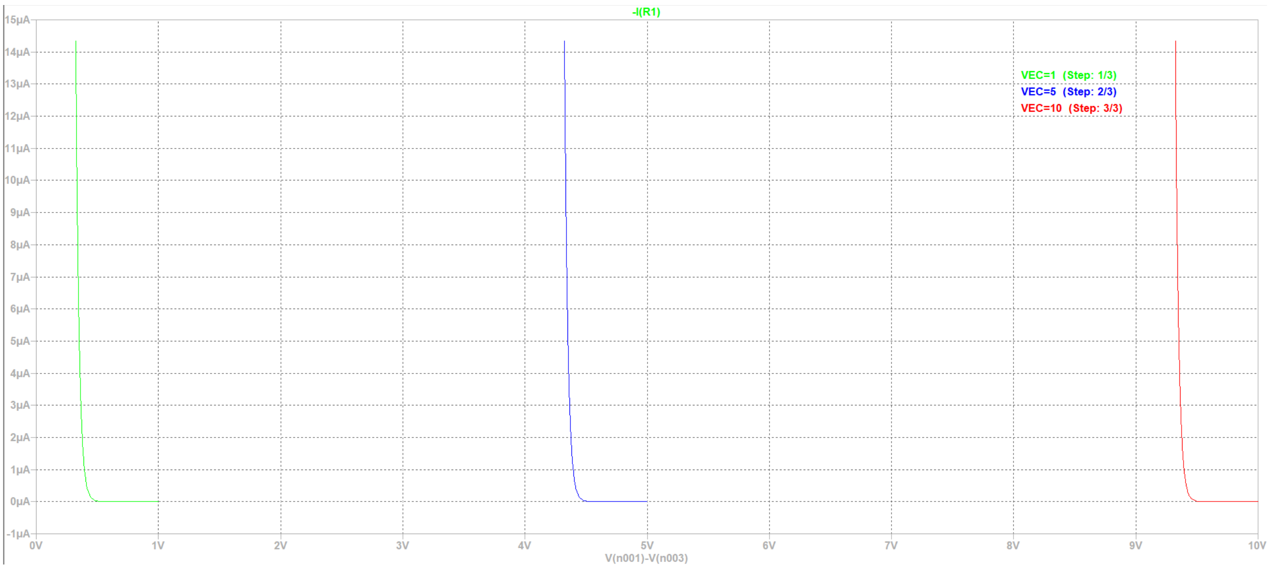
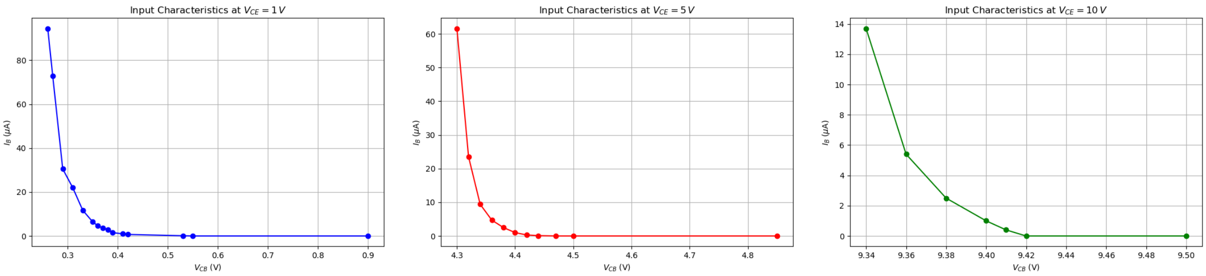


Figure 19: LTspice Simulation of CC Input Characteristics (I_B vs V_{CB})

8.6 Experimental Characteristic Plots



(a) $V_{CE} = 1V$ (b) $V_{CE} = 5V$ (c) $V_{CE} = 10V$

Figure 20: Experimental plots of CC Input characteristics (I_B vs V_{CB})

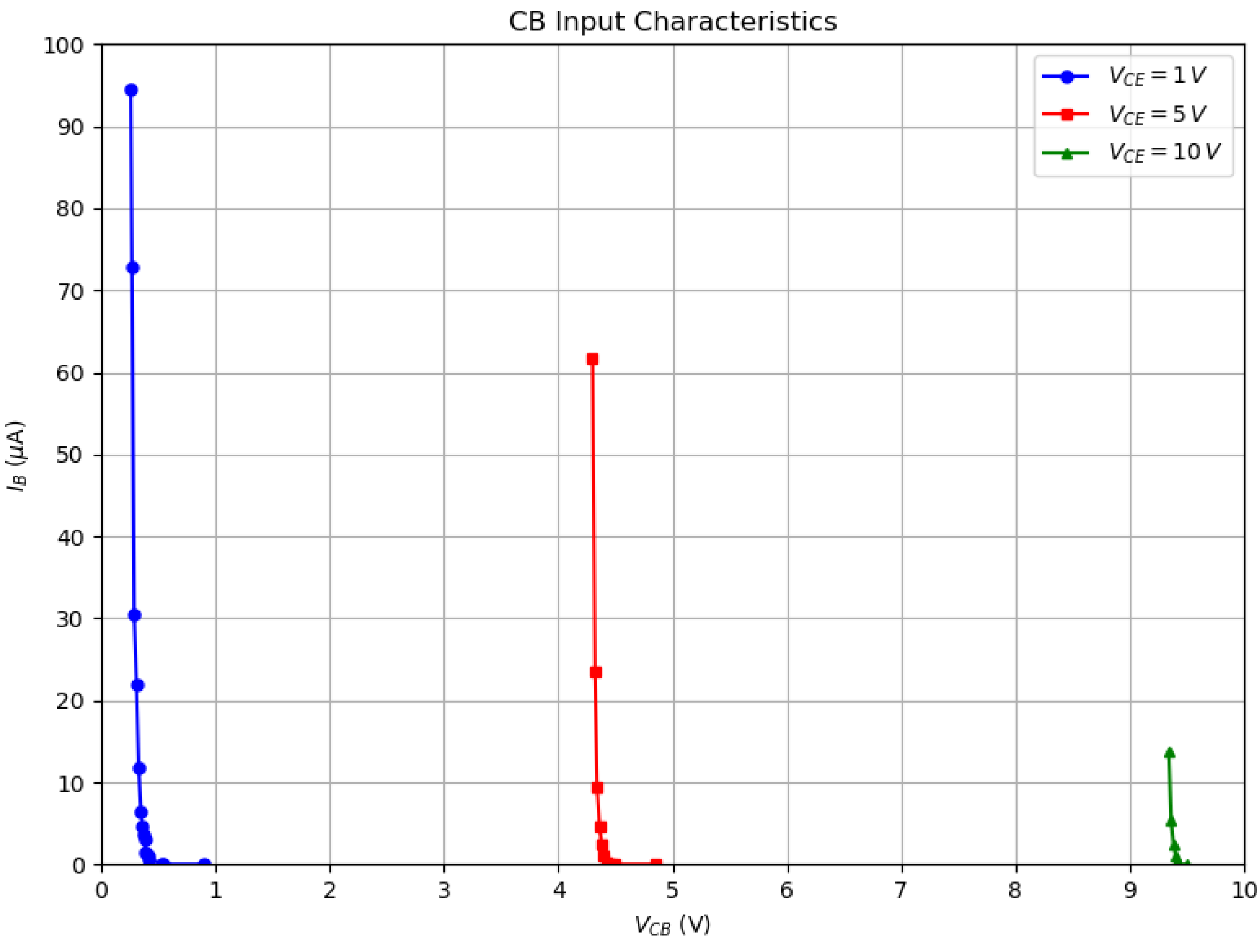


Figure 21: CC Input Characteristics I_B vs V_{CB}

8.7 Calculations

8.7.1 Input Dynamic Resistance (r_i)

In the Common Collector (CC) configuration with constant V_{CE} (where $V_{EB} = V_{EC} - V_{CB}$), the dynamic input resistance r_i is defined by the reciprocal slope of the I_B versus

V_{CB} curve in the active conduction region:

$$r_i = \left(\frac{\Delta V_{BE}}{\Delta I_B}\right)_{V_{CE}=\text{constant}}$$

1. **At $V_{CE} = 1\,\text{V}$:**

   Taking the active conduction interval from $V_{CB} = 0.26\,\text{V}$ to $0.33\,\text{V}$:

$$\begin{aligned}
\Delta V_{CB} &= 0.33\,\text{V} - 0.26\,\text{V} = 0.07\,\text{V} \\
\Delta I_B &= 94.5\,\mu\text{A} - 11.7\,\mu\text{A} = 82.8\,\mu\text{A} = 82.8 \times 10^{-6}\,\text{A} \\
\boldsymbol{r_{i1}} &= \frac{0.07\,\text{V}}{82.8 \times 10^{-6}\,\text{A}} = 845.41\,\Omega \quad (\approx 0.85\,\text{k}\Omega)
\end{aligned}$$

2. **At $V_{CE} = 5\,\text{V}$:**

   Taking the active conduction interval from $V_{CB} = 4.30\,\text{V}$ to $4.34\,\text{V}$:

$$\begin{aligned}
\Delta V_{CB} &= 4.34\,\text{V} - 4.30\,\text{V} = 0.04\,\text{V} \\
\Delta I_B &= 61.6\,\mu\text{A} - 9.4\,\mu\text{A} = 52.2\,\mu\text{A} = 52.2 \times 10^{-6}\,\text{A} \\
\boldsymbol{r_{i2}} &= \frac{0.04\,\text{V}}{52.2 \times 10^{-6}\,\text{A}} = 766.28\,\Omega \quad (\approx 0.77\,\text{k}\Omega)
\end{aligned}$$

3. **At $V_{CE} = 10\,\text{V}$:**

   Taking the active conduction interval from $V_{CB} = 9.34\,\text{V}$ to $9.38\,\text{V}$:

$$\begin{aligned}
\Delta V_{CB} &= 9.38\,\text{V} - 9.34\,\text{V} = 0.04\,\text{V} \\
\Delta I_B &= 13.7\,\mu\text{A} - 2.5\,\mu\text{A} = 11.2\,\mu\text{A} = 11.2 \times 10^{-6}\,\text{A} \\
\boldsymbol{r_{i3}} &= \frac{0.04\,\text{V}}{11.2 \times 10^{-6}\,\text{A}} = 3.57\,\text{k}\Omega
\end{aligned}$$

8.8 Result

The CC input characteristics were plotted for $V_{CE} = 1\,\text{V}$, $5\,\text{V}$, and $10\,\text{V}$.

The input dynamic resistance obtained from the measured characteristics is

$$r_i = 845.41\,\Omega \quad (V_{CE} = 1\,\text{V}),$$

$$r_i = 766.28\,\Omega \quad (V_{CE} = 5\,\text{V}),$$

and

$$r_i = 3.57\,\text{k}\Omega \quad (V_{CE} = 10\,\text{V}).$$

8.9 Observations

- The base current I_B decreases with increase in V_{CB} for a fixed value of V_{CE}.
- The input characteristics show nonlinear, diode-like behaviour in the conduction region.
- The characteristics shift with change in the fixed value of V_{CE}.
- The input dynamic resistance is determined from the slope of the characteristic in the active conduction region.
- The input resistance varies for different values of V_{CE}.

9 Verification of the α–β–γ Relationships

The theoretical relationships between the current gains of a BJT are

$$\beta = \frac{\alpha}{1-\alpha}$$

and

$$\gamma = \beta + 1.$$

These relationships are verified using the experimentally determined values of α, β, and γ obtained from the CB, CE, and CC configurations.

9.1 Verification of the α–β Relationship

From the CB configuration, the experimentally determined current gain is

$$\alpha_{\text{exp}} = 0.9974.$$

The corresponding value of β calculated using the theoretical relationship is

$$\beta_\alpha = \frac{\alpha_{\text{exp}}}{1-\alpha_{\text{exp}}}$$

$$\beta_\alpha = \frac{0.9974}{1 - 0.9974} = 383.6.$$

From the CE configuration, the experimentally determined current gain is

$$\beta_{\text{exp}} = 350.75.$$

The percentage difference between the experimentally measured and calculated values is

$$\%\ \text{difference} = \frac{|\beta_{\text{exp}} - \beta_\alpha|}{\beta_{\text{exp}}} \times 100$$

$$\%\ \text{difference} = \frac{|350.75 - 383.6|}{350.75} \times 100 = \mathbf{9.4\%}.$$

Thus, the experimentally determined values of α and β show reasonable agreement with the theoretical relationship, considering experimental and measurement uncertainties.

9.2 Verification of the β–γ Relationship

The theoretical relationship between the current gains of the Common Emitter and Common Collector configurations is

$$\gamma = \beta + 1.$$

From the CE configuration, the experimentally determined current gain is

$$\beta_{\text{exp}} = 350.75.$$

Therefore, the theoretical value of γ calculated from the measured β is

$$\gamma_\beta = \beta_{\text{exp}} + 1 = 350.75 + 1 = \mathbf{351.75}.$$

From the CC configuration, the experimentally determined current gains at $V_{CE} = 4\,\text{V}, 6\,\text{V}$, and $8\,\text{V}$ are

$$\gamma_4 = 320.43, \qquad \gamma_6 = 342.14, \qquad \gamma_8 = 359.11.$$

The average experimental current gain is

$$\gamma_{\text{avg}} = \frac{\gamma_4 + \gamma_6 + \gamma_8}{3}$$

$$\gamma_{\text{avg}} = \frac{320.43 + 342.14 + 359.11}{3} = \mathbf{340.56}.$$

The percentage error between the theoretical and experimental values is

$$\%\ \text{error} = \frac{|\gamma_\beta - \gamma_{\text{avg}}|}{\gamma_\beta} \times 100$$

$$\%\ \text{error} = \frac{|351.75 - 340.56|}{351.75} \times 100 = \mathbf{3.18\%}.$$

Thus, the experimentally determined average value of γ shows reasonable agreement with the theoretical value obtained from β, with a percentage error of **3.18%**.

9.3 Observations

The experimentally determined values of α, β, and γ are in reasonable agreement with the theoretical relationships

$$\beta = \frac{\alpha}{1-\alpha}$$

and

$$\gamma = \beta + 1.$$

The deviations from the theoretical relationships can be attributed to transistor parameter variations, measurement uncertainties, and the different operating conditions used in the experimental measurements.

10 Precautions

1. The emitter, base, and collector terminals of the BC547 should be identified correctly before making the circuit connections.
2. The power supply should be switched off before changing any circuit connections.
3. Ammeters must always be connected in series, while voltmeters must be connected in parallel.
4. The polarity of the meters and power supply should be checked before taking readings.
5. The maximum voltage and current ratings of the BC547 transistor should not be exceeded.

6. The required constant current or voltage should be maintained accurately while taking the readings.
7. Loose connections on the breadboard should be avoided.
8. Readings should be taken only after the meters have stabilized.
9. The slopes used for calculating r_i and r_o should be taken from the appropriate active-region portion of the characteristics.
10. Average values of r_i or r_o should not be used for calculating current gain unless specifically required; gain should be calculated using the corresponding characteristic and operating points.

11 Overall Results

The important experimental parameters obtained for the three BJT configurations are summarized below.

- **Common Emitter (CE):** The experimental current gain was $\beta_{\text{exp}} = 350.75$. The input dynamic resistance was found to be $5.26\,\text{k}\Omega$, $4.65\,\text{k}\Omega$, and $2.20\,\text{k}\Omega$ for $V_{CE} = 1\,\text{V}$, $5\,\text{V}$, and $10\,\text{V}$, respectively. The output dynamic resistance was approximately $12.5\,\text{k}\Omega$.
- **Common Base (CB):** The experimental current gain was $\alpha_{\text{exp}} = 0.9974$. The input dynamic resistance was found to be $727.27\,\Omega$, $645.16\,\Omega$, and $666.67\,\Omega$ for $V_{CE} = 1\,\text{V}$, $5\,\text{V}$, and $10\,\text{V}$, respectively. The output dynamic resistance was approximately $1.20\,\text{M}\Omega$.
- **Common Collector (CC):** The average experimental current gain was $\gamma_{\text{avg}} = 340.56$. The input dynamic resistance was found to be $845.41\,\Omega$, $766.28\,\Omega$, and $3.57\,\text{k}\Omega$ for $V_{CE} = 1\,\text{V}$, $5\,\text{V}$, and $10\,\text{V}$, respectively. The output dynamic resistance was $14.71\,\text{k}\Omega$, $1.78\,\text{k}\Omega$, and $1.34\,\text{k}\Omega$ for the corresponding operating conditions.

12 Conclusion

The input and output characteristics of the BC547 transistor were studied experimentally in the CE, CB, and CC configurations. The obtained characteristics were in good agreement with the expected transistor behaviour.

- **Common Emitter (CE):** The CE configuration exhibited significant current gain and a comparatively high input and output resistance in the active region.

- **Common Base (CB):** The CB configuration exhibited a current gain close to unity, low input resistance, and high output resistance.
- **Common Collector (CC):** The CC configuration exhibited high current gain, low output resistance, and characteristics suitable for impedance matching.
- **Gain Relationships:** The experimentally determined values showed reasonable agreement with the theoretical relationships $\beta = \frac{\alpha}{1-\alpha}$ and $\gamma = \beta + 1$.

Overall, the experimental characteristics and calculated parameters were found to be in good agreement with the theoretical behaviour of the BJT. Minor deviations were attributed to measurement uncertainties and practical circuit conditions.